Conformal Prediction on MNIST
MNIST

An R re-implementation of LAC, Adaptive Prediction Sets and the evaluation protocol
evaluation protocol

PAPER

Angelopoulos & Bates, 2021 — Sections 1, 2.1, 4

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SCRIPT

Conformal_Prediction_MNIST_V6.4.R

FRAMING

Paper, Scope and Deliverables

WHAT WE RE-IMPLEMENTED

Section 1 — split conformal prediction with the least-ambiguous-classifier score.

Section 2.1 — adaptive prediction sets built on cumulative softmax mass.

Section 4 — coverage and set-size checks, repeated over random splits.

WHAT WE LEFT OUT

Conformalized quantile regression (2.2), scalar uncertainty estimates (2.3), conformalized Bayes (2.4) and risk-controlling prediction sets (3) — all outside the classification setting we chose.

BASE MODEL

The neural network is adapted from public code, not written by us. It is deliberately treated as a black box — which is exactly the setting conformal prediction is designed for.

METHOD

Split Conformal Prediction in Four Steps

01

Hold out calibration data

Data the fitted model never saw. We use 2,000 images.

02

Define a score $s(X, Y)$

LAC: $1 - p(\text{true class})$. APS: cumulative mass down to the true class.

03

Take the conformal quantile

$\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration score, with $\alpha = 0.10$.

04

Form the set $T(X)$

Every label whose score falls at or below $\hat{q}$. Coverage $\geq 1 - \alpha$ follows.

SETUP

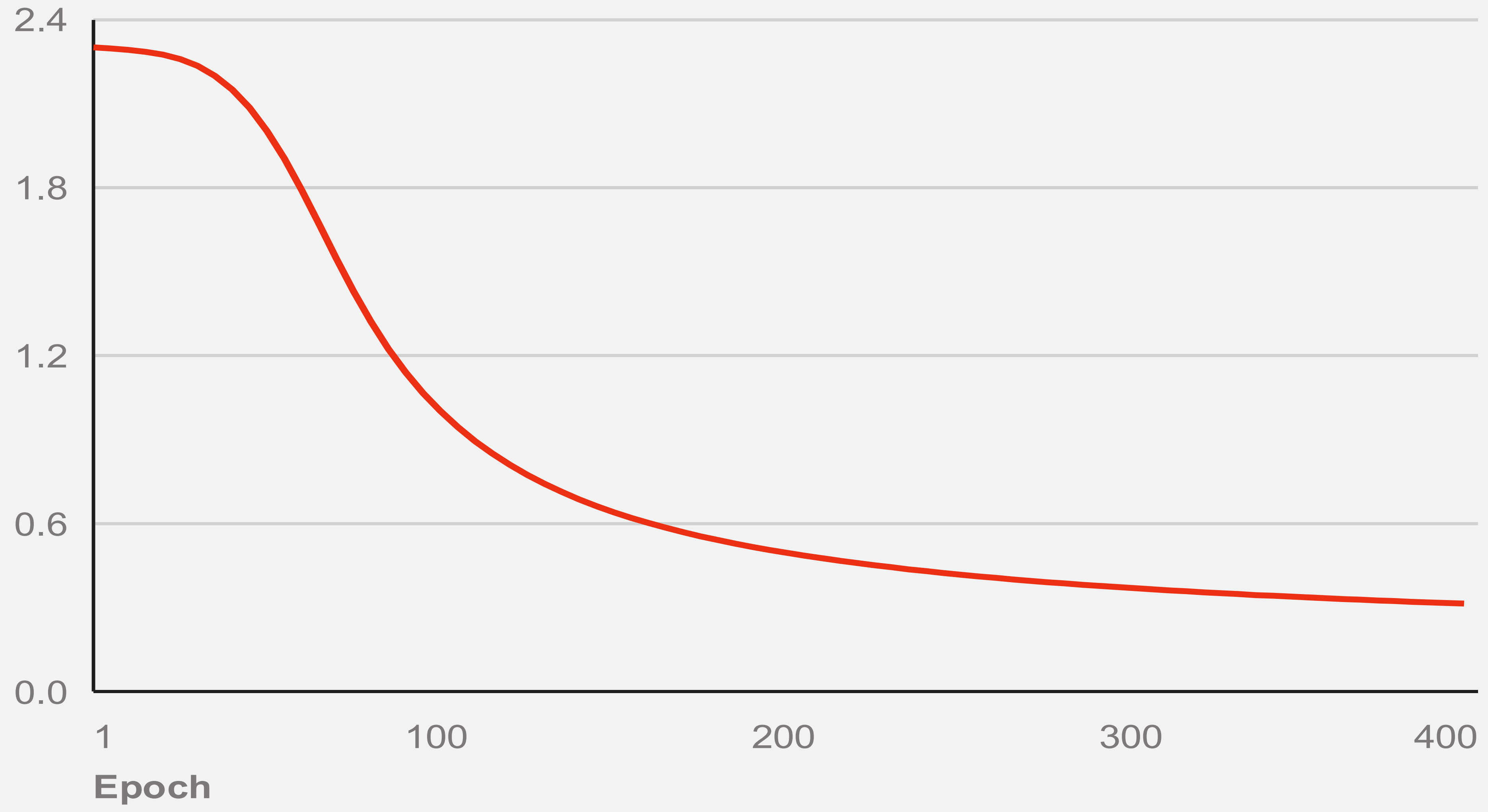
Data, Splits and Base Classifier

TRAIN	CALIBRATE	TEST
4,000	2,000	2,000

784 pixels scaled to [0, 1] → one hidden layer of 64 ReLU units → softmax over 10 digits. Full-batch gradient descent, learning rate 0.1, 400 epochs, seed fixed.

Accuracy, training set	**91.95%**
Accuracy, test set	**89.10%**

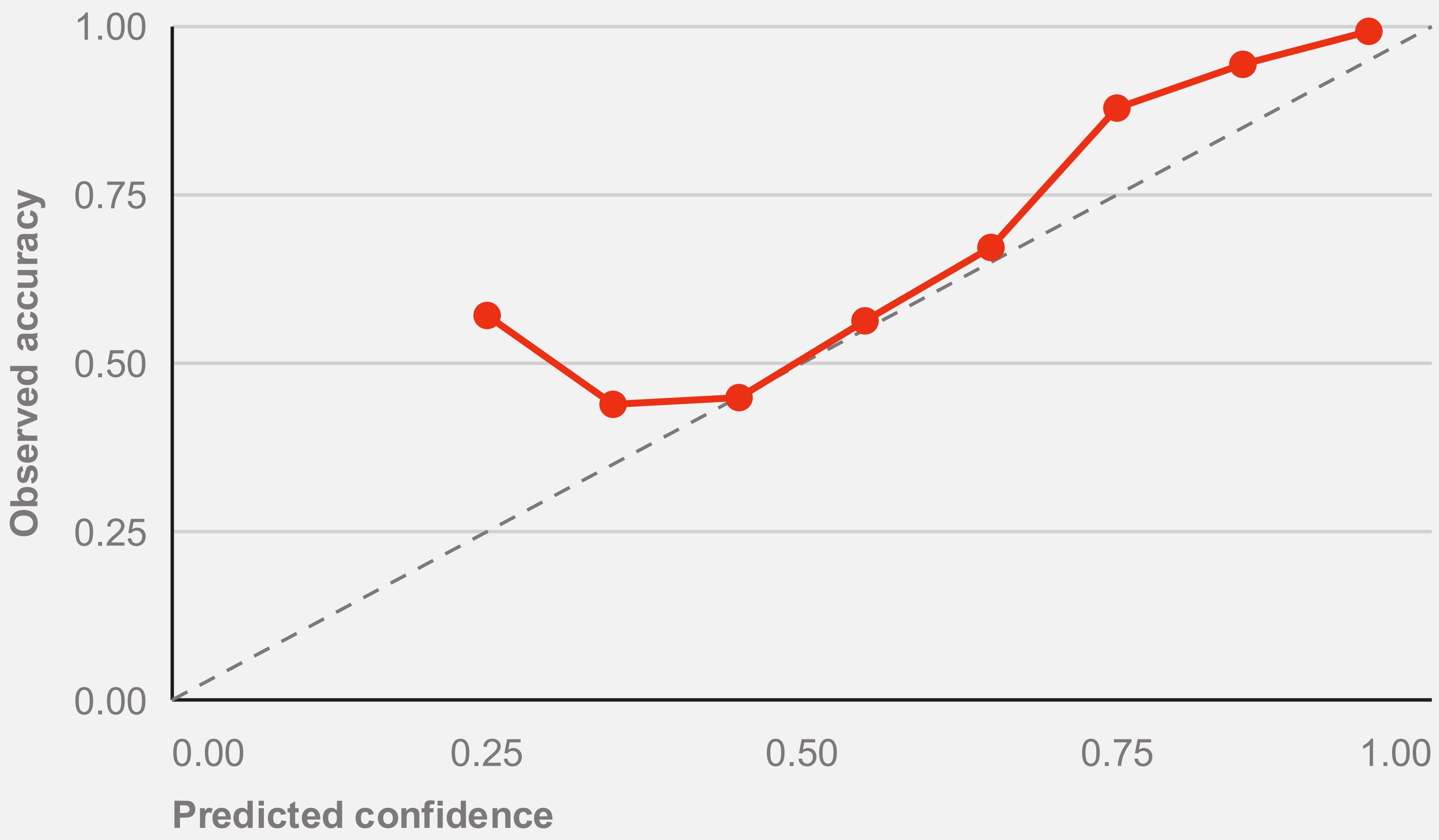
CROSS-ENTROPY LOSS OVER 400 EPOCHS



Loss falls from 2.302 to 0.312. The curve is still descending — the classifier is adequate, not tuned.

DIAGNOSTIC

Calibration of the Raw Softmax



Observed accuracy against predicted confidence, ten bins, 2,000 test images. Dashed line is perfect calibration.

The softmax is not a probability we can trust as it stands.

In the 0.75 bin the network is right 87.9% of the time and in the 0.85 bin 94.4% — it understates its own accuracy. Below 0.5 the estimates wander and the bins are thin.

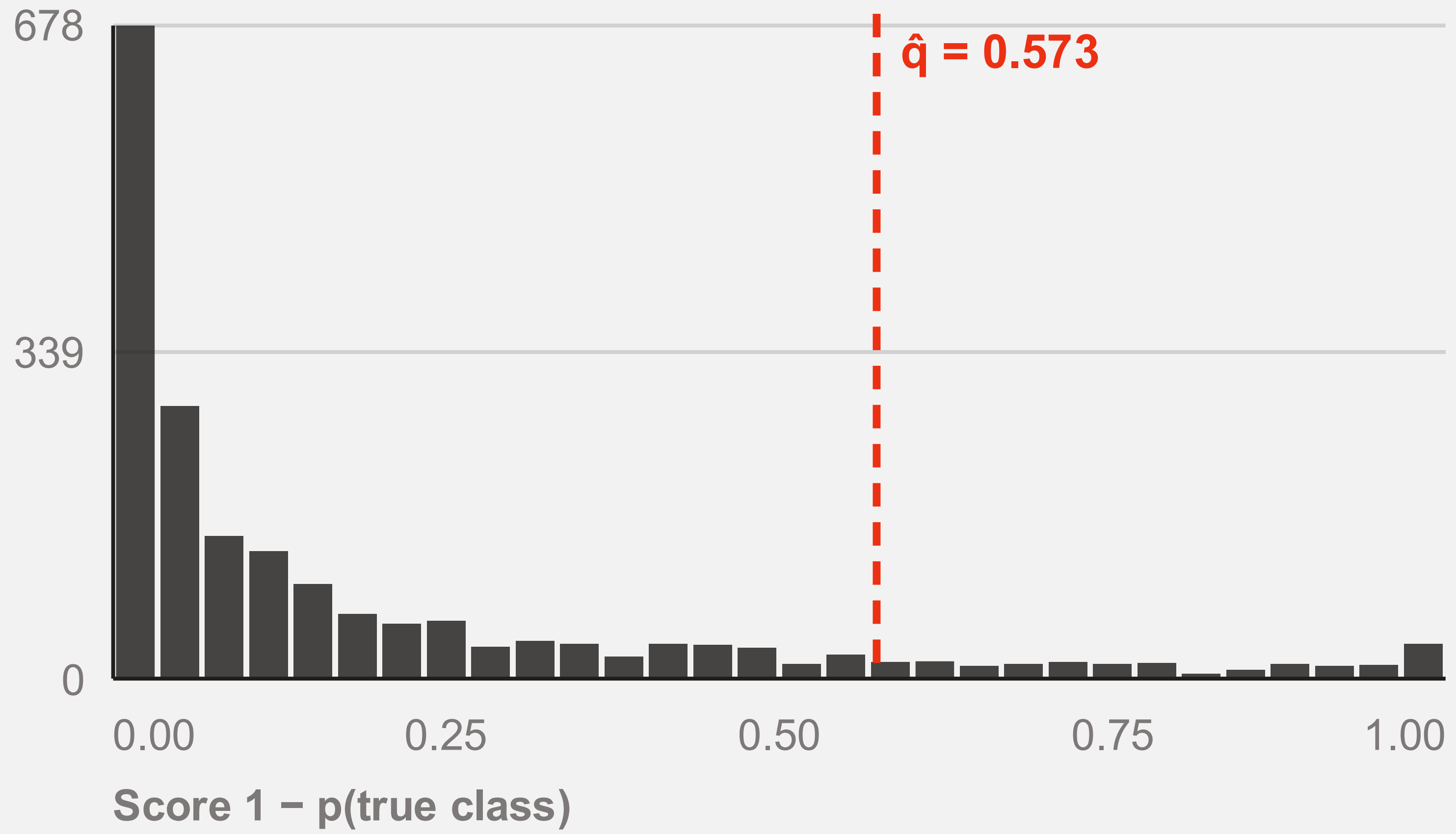
55% of test images arrive with confidence above 0.90, where accuracy is 99.3%. The remaining tail is where prediction sets have to do the work.

This is why the paper insists the guarantee is distribution-free: nothing above needs to be fixed for conformal prediction to be valid.

CALIBRATION

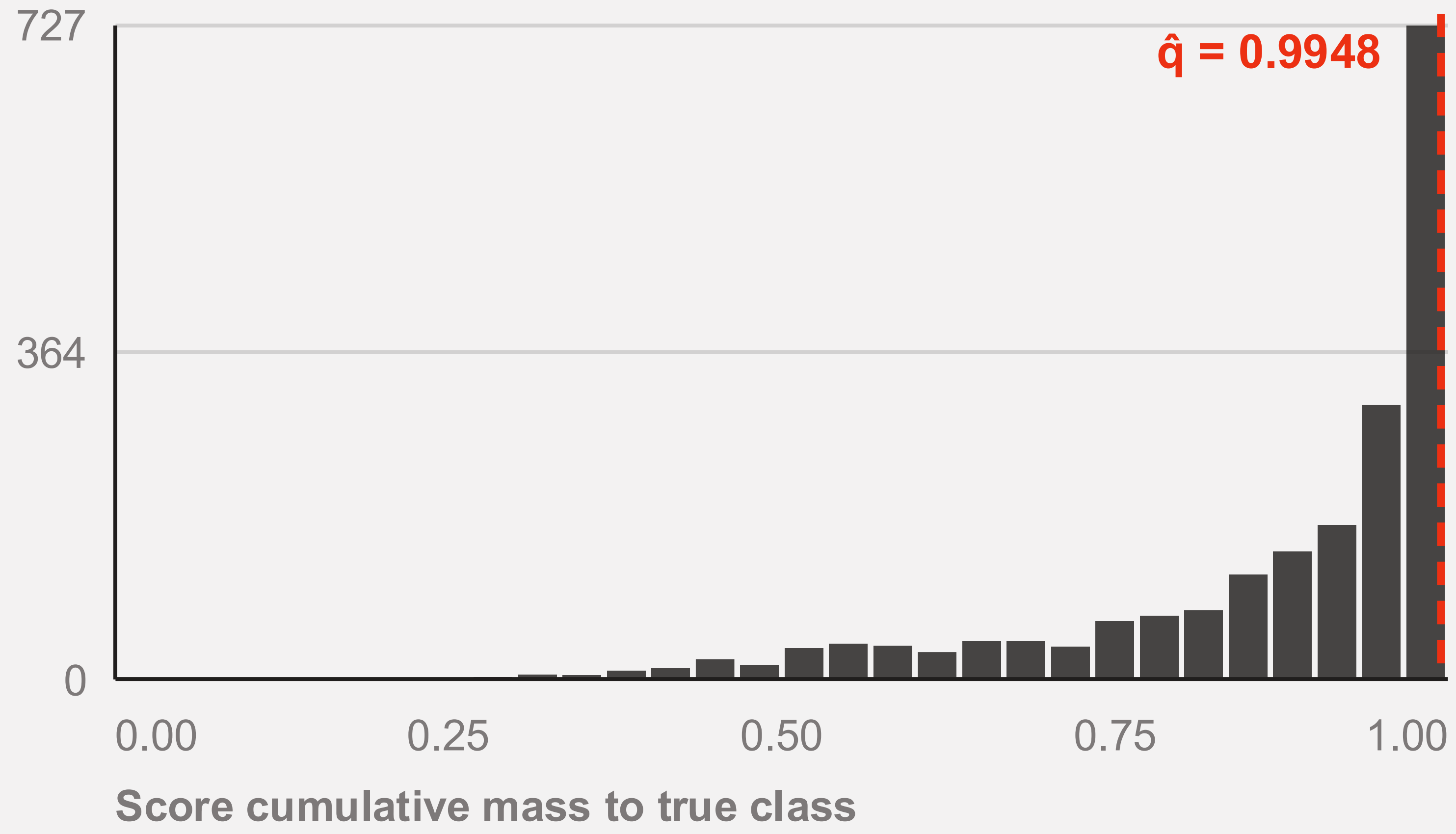
Conformal Scores and Thresholds

LAC — s = 1 − p(true class)



Heavily right-skewed: 48% of calibration images score below 0.033. The 90th percentile lands at **$\hat{q}$ = 0.573** , so a label enters the set when its softmax exceeds 0.427.

APS — cumulative mass to true class



Mirror image: mass concentrates just under 1. The quantile is pushed to **$\hat{q}$ = 0.9948** , which forces sets to accumulate almost all the probability mass.

RESULTS — SECTION 1

LAC Results on the Test Split

THRESHOLD Q	EMPIRICAL COVERAGE	MEAN SET SIZE
0.573	88.15%	0.98

Sets are almost always a single digit: 96.1% of test images get exactly one label.

Only 18 images (0.9%) receive two or more labels. LAC is the most efficient score in the paper, and this is what that efficiency looks like.

60 test images (3.0%) receive an **empty set**.

When no softmax value exceeds $1 - \hat{q} = 0.427$, the rule returns nothing and coverage on those images is zero by construction. Marginal coverage still holds; the sets are simply useless where the model is most confused.

RESULTS — SECTION 2.1

APS Results on the Test Split

THRESHOLD Q	EMPIRICAL COVERAGE	MEAN SET SIZE
0.995	99.70%	4.04

The guarantee is a lower bound, and APS sits far above it.

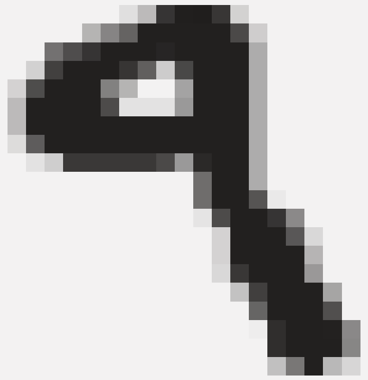
We implemented Equation 2 without the randomised tie-breaking the paper mentions in its footnote. The greedy rule always adds one class past the threshold, so every set is at least as large as it needs to be — and never empty.

Overconfident tails inflate $\hat{q}$.

For the ~11% of calibration images the network gets badly wrong, the true class sits near the bottom of the ranking, so the cumulative mass needed to reach it is close to 1. The 90th percentile of that distribution is 0.9948, and every test image then pays for it.

ILLUSTRATION

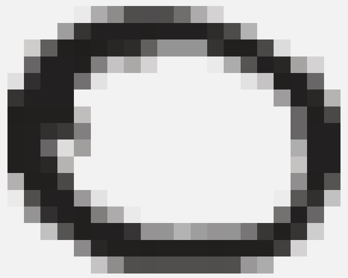
Prediction Sets on Real Digits



TRUE 9

LAC {9}

APS {3,4,5,8,9}



TRUE 0

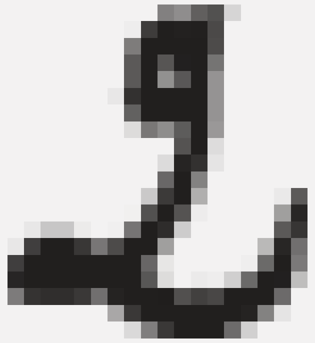
LAC {0}

APS {0,2,6}

TRUE 7

LAC {7}

APS {7,9}



TRUE 2

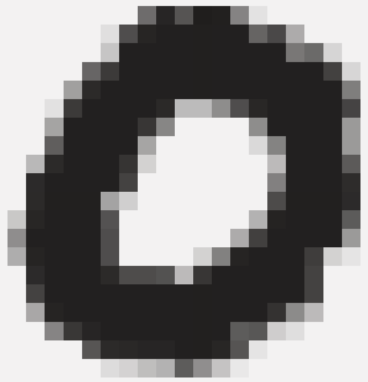
LAC {2}

APS {2,6}

TRUE 6

LAC {6}

APS {2,5,6,8}



TRUE 0

LAC {0}

APS {0}

Both methods contain the truth in all six cases. APS spends extra labels to say where the image is ambiguous; LAC says nothing about it.

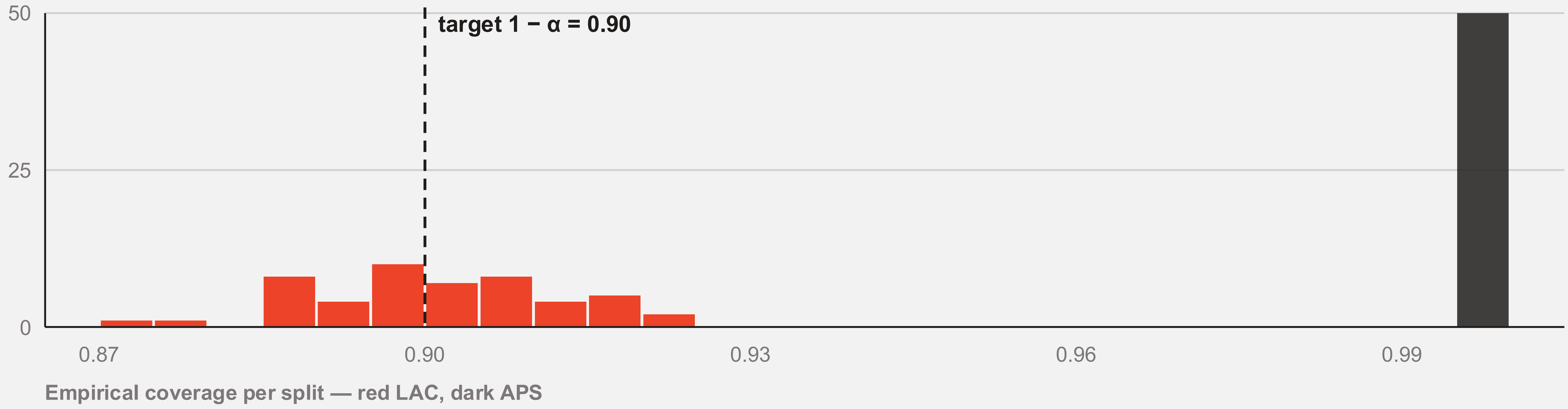
COMPARISON

LAC versus APS, Side by Side

METRIC	LAC — SECTION 1	APS — SECTION 2.1	READING
Threshold $\hat{q}$	0.573	0.995	Different scales, not comparable directly
Coverage, 50 splits	90.08%	99.71%	LAC hits the target; APS over-covers
Mean set size	1.00	4.08	Four times the labels for the same nominal α
Single-label sets	96.1%	9.5%	LAC behaves like a point classifier
Empty sets	3.0%	0.0%	LAC abstains silently on hard images
Worst digit coverage	79.4%	99.1%	Conditional coverage is where LAC pays

EVALUATION — SECTION 4.1

Coverage over Fifty Random Splits



LAC COVERAGE

90.08%

σ = 1.12 pts, range 87.4 – 92.1

APS COVERAGE

99.71%

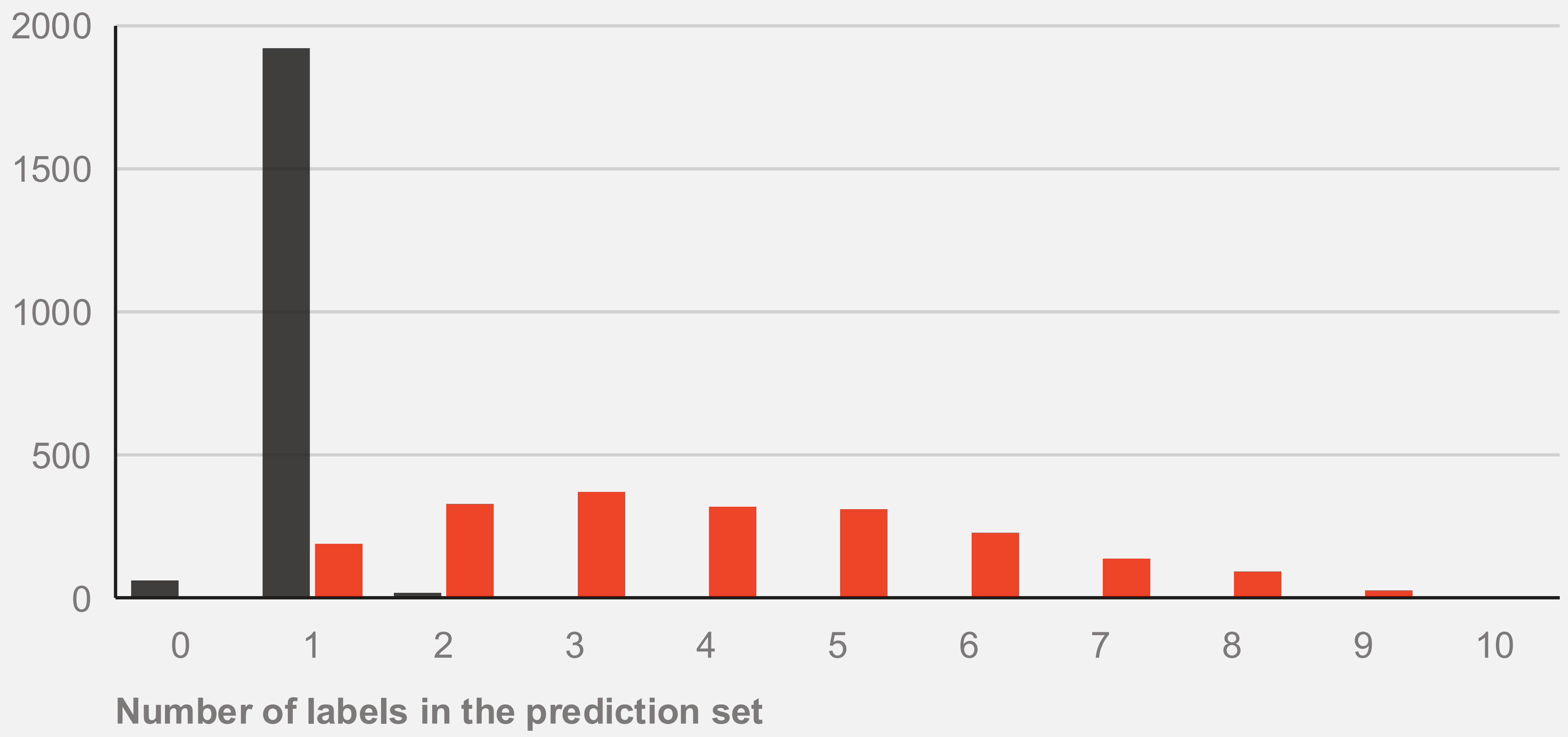
σ = 0.10 pts, range 99.5 – 99.9

INTERPRETATION

The 88.15% seen on the single split was ordinary calibration noise. Averaged over splits the guarantee is met almost exactly — the theorem is about the average over calibration sets, not about one draw.

EVALUATION — SECTION 4.1

Set Size Distributions and Adaptiveness



Number of test images by set size. Dark bars LAC, red bars APS.

LAC produces a spike; APS produces a distribution.

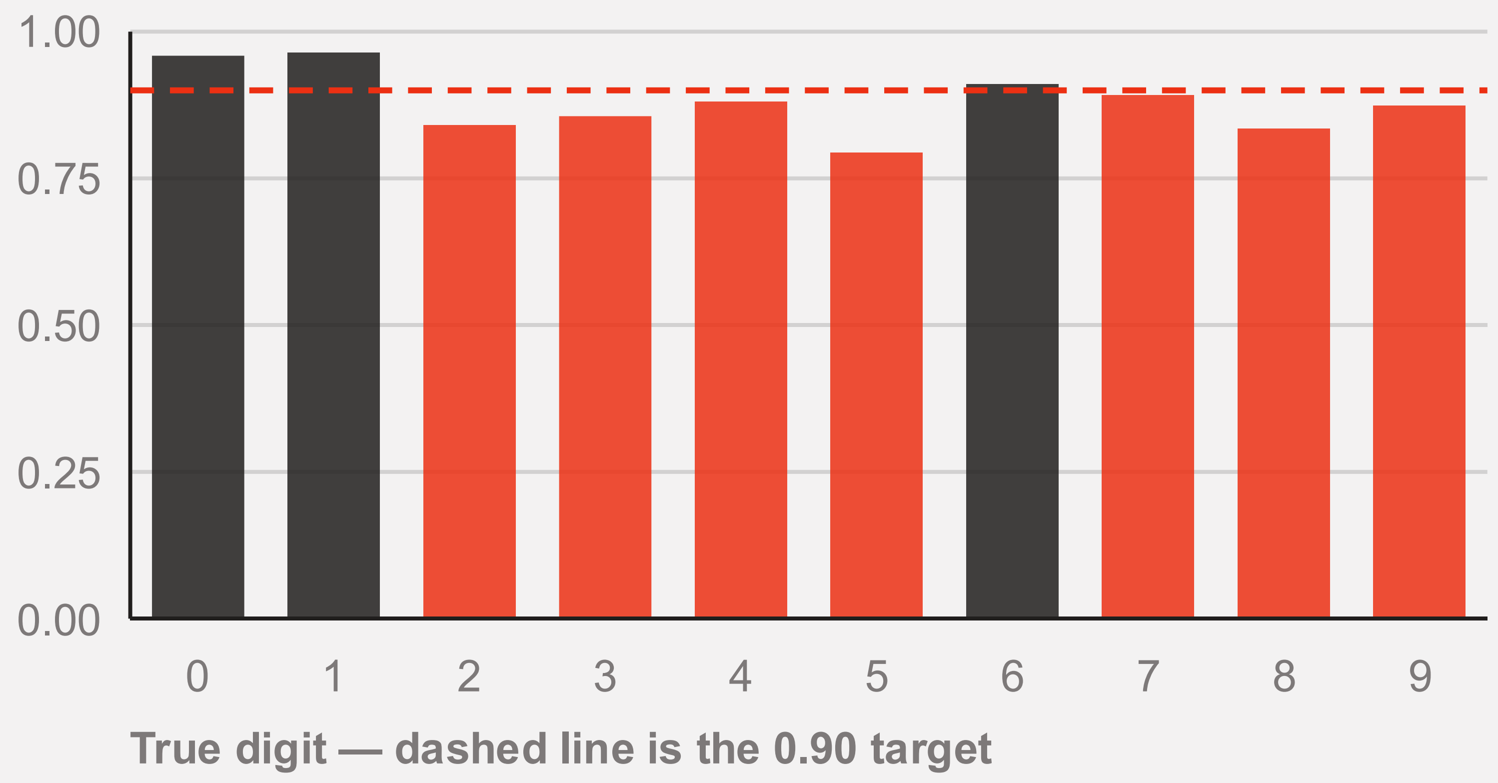
The paper argues a wider spread is desirable: it means the procedure separates easy inputs from hard ones. APS spans one to nine labels, LAC only zero to two.

But spread is not free. APS pays 4.08 labels per image on average, and its narrowest sets still over-cover. Adaptiveness here comes bundled with inefficiency, not instead of it.

EVALUATION — SECTION 4.2

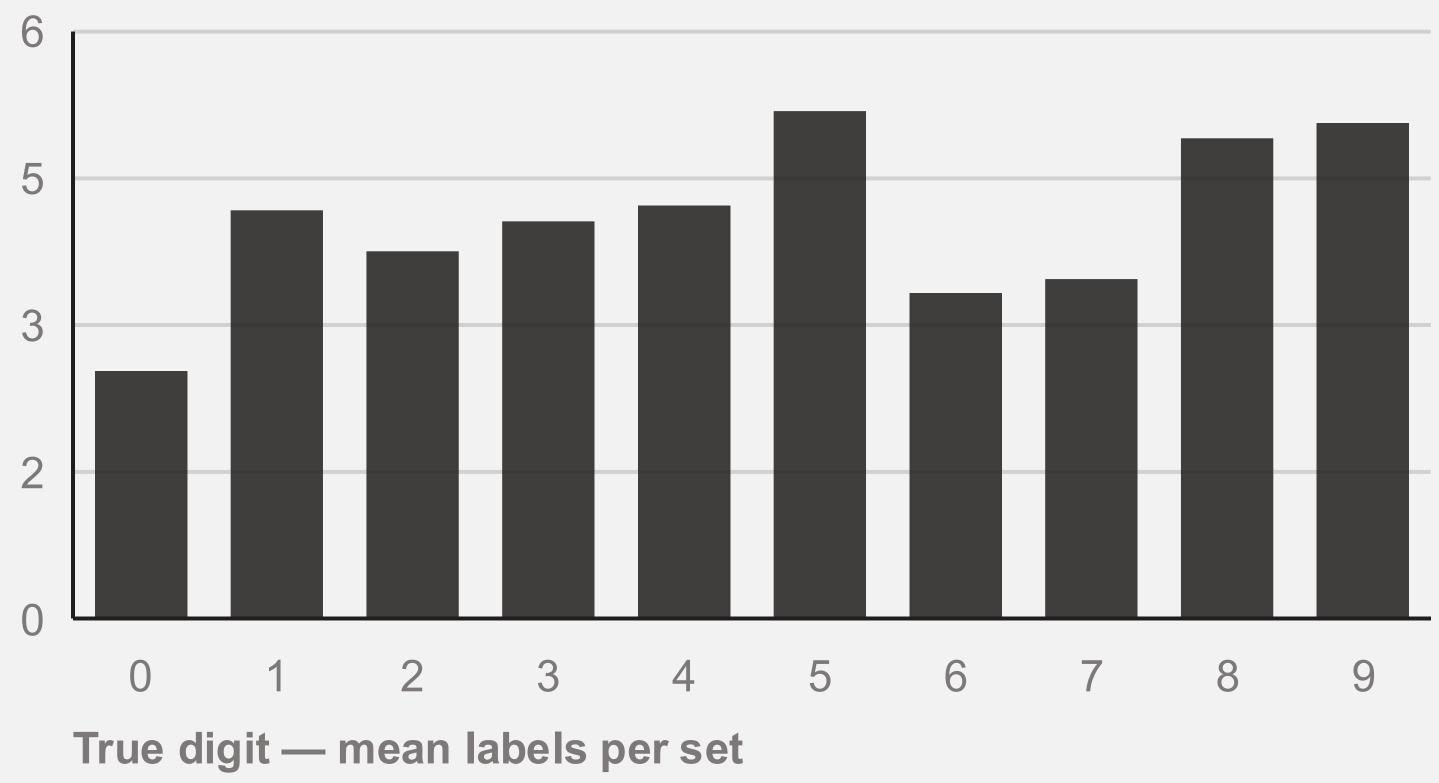
Coverage by Digit — Feature-Stratified

LAC coverage per digit



FSC metric = **0.794** on the digit 5, against a 0.90 target. Zeros and ones are over-covered at 0.96.

APS mean set size per digit



APS keeps every digit above 0.99 coverage and moves the cost into size: 2.53 labels for zeros, 5.19 for fives.

EVALUATION — SECTION 4.2

Coverage by Set Size — Set-Stratified

LAC — SSC = 0.000

size 0 · 60 images	0.000
size 1 · 1,922 images	0.909
size 2+ · 18 images	0.833

Marginal coverage is bought by covering the easy 96% and abandoning the rest.

APS — SSC = 0.995

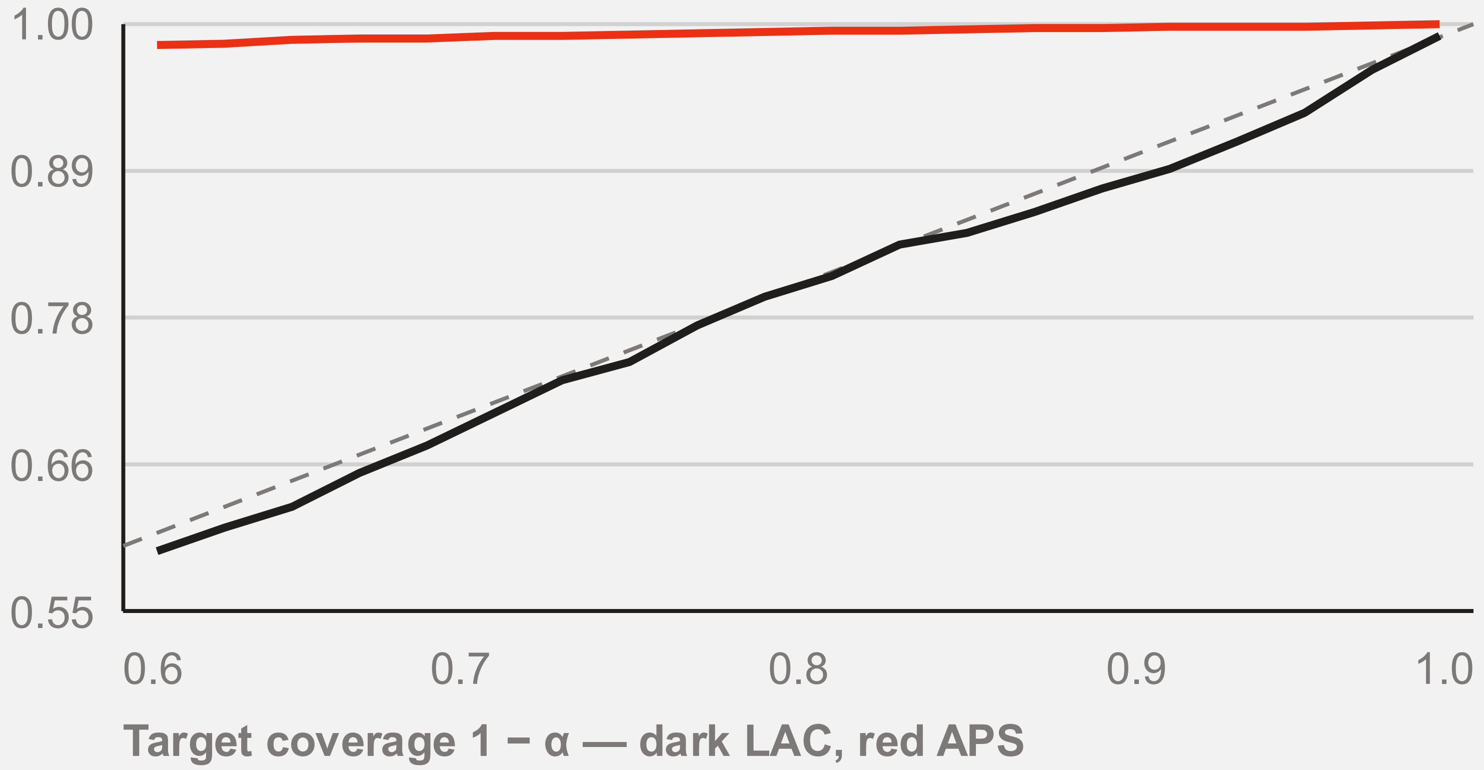
size 1 · 189 images	1.000
size 2 · 329 images	0.997
size 3–4 · 689 images	0.999
size 5+ · 793 images	0.995

Uniform across bins — but uniformly above target, which is its own kind of miscalibration.

SENSITIVITY

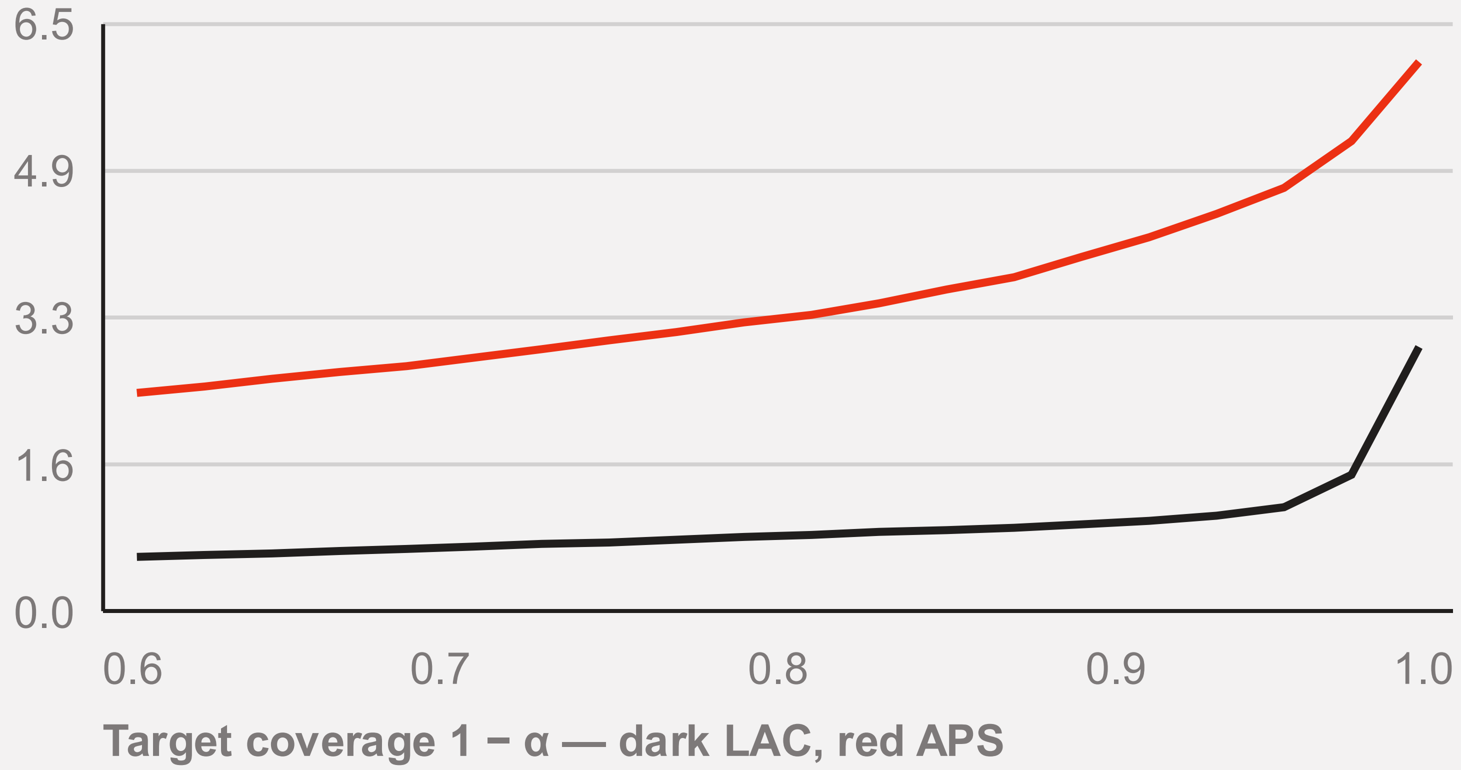
The Coverage–Efficiency Trade-off

Empirical against target coverage



LAC follows the diagonal across the whole sweep, one to two points low. APS is pinned near 1.00 regardless of what we ask for.

Mean set size against target



The price of certainty is convex: LAC needs 1.00 labels for 91% and 2.93 for 99%. APS starts at 2.42 and reaches 6.08.

CONCLUSIONS

Findings, Limitations and Next Steps

The coverage guarantee reproduced on the first try. Everything difficult was about which score to use.

FINDING 01

LAC met the 90% target over 50 splits with a mean set size of 1.00 — but under-covered fives at 79% and returned empty sets for 3% of images.

FINDING 02

APS equalised coverage across digits and set sizes, at four times the labels and a 9.7-point overshoot — validity is easy, tightness is not.

FINDING 03

Single-split coverage was 88.15%; the fifty-split average was 90.08%. Reporting one split would have looked like a failed reproduction.

LIMITATIONS

8,000 of 70,000 images, one network, one seed. APS omits the randomised tie-breaking that would remove most of its overshoot.

NEXT STEPS

Add randomised APS, report Mondrian per-digit calibration for LAC, and re-run the sweep on a convolutional base model.